
Hollow Rectangular Waveguide Modal Analysis

Square-Guide Degeneracy, Cutoff, and Multimode CST
Interpretation at 10 GHz

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Independent Engineering Portfolio Project

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Portfolio Context and Evidence

Project Context. This work originated from a graduate-level antenna engineering simulation completed during Summer 2026. The present document is an independently prepared public portfolio edition intended to demonstrate guided-wave analysis, CST full-wave simulation, multimode interpretation, numerical verification, and technical documentation.

Public Portfolio Edition. The original academic report has been reorganized and edited for professional and educational use. Course-specific instructions, grading material, student-identification information, institutional branding, instructor information, and administrative content have been removed. This document does not represent an official publication or endorsement by any university, instructor, software vendor, or other institution.

Evidence Classification. The results presented in this report are based on closed-form rectangular-waveguide calculations, an executable Python verification, and CST Studio Suite simulations. No waveguide hardware was fabricated, and no VNA, field-probe, or antenna-range measurement campaign was conducted. The reported cutoff frequencies, impedances, scattering parameters, fields, and surface-current patterns are therefore classified as analytical or simulated evidence, not measured performance.

Simulation Scope. The model is an ideal vacuum-filled, PEC-walled, uniform square waveguide. The public package includes selected CST figures but not the native binary model or raw exported field and S-parameter datasets. Formal mesh, port-basis, and calibration-plane convergence studies were not completed. Finite conductivity, manufacturing tolerances, transitions, and a horn flare remain outside the present scope and are discussed explicitly as limitations.

Prepared as part of the Hossein Molhem Engineering Portfolio.

Abstract

This report presents an analytical and CST Studio Suite study of a hollow square waveguide intended as a modal feed-section case study for horn-antenna applications. The modeled interior is $20\text{ mm} \times 20\text{ mm}$ with a length of 90 mm , vacuum filling, and ideal PEC walls. Two waveguide ports were solved with three modes over $8\text{--}15\text{ GHz}$, with field evaluation centered at 10 GHz .

Closed-form theory predicts degenerate TE_{10} and TE_{01} cutoff frequencies of 7.4948 GHz , followed by a $(1,1)$ -family cutoff near 10.5993 GHz . At 10 GHz , the first two modes propagate while the third solved mode is slightly below cutoff. The analytical dominant-mode guide wavelength is 45.284 mm , the TE wave impedance is $569.06\ \Omega$, and the 90 mm section is approximately 1.987 guide wavelengths long.

CST reports Port 1 cutoff frequencies of 7.490707 , 7.491242 , and 10.5935 GHz , and nearly identical values at Port 2. The relative differences from theory remain below 0.061% . Multimode S-parameter plots show very small same-mode reflection and near-unity transmission into the alternate member of the degenerate output basis. This is not physical mode conversion in a discontinuity; it reflects the non-unique polarization basis of a perfectly square guide. Field and surface-current plots support propagating behavior for the first two modes and localized near-cutoff behavior for the third.

The model is intentionally idealized. No mesh-convergence campaign, finite-conductivity sensitivity study, transition model, horn flare, prototype, or measurement is included. Accordingly, the project demonstrates modal reasoning and simulation interpretation rather than a production-ready horn-feed design.

Project Objective

The objective is to establish and verify the modal behavior of an ideal hollow waveguide with a square $20\text{ mm} \times 20\text{ mm}$ cross section and a physical length of 90 mm . The study focuses on operation at 10 GHz over an $8\text{--}15\text{ GHz}$ simulation band.

The work addresses five engineering questions:

1. Which low-order TE and TM modes are above cutoff at 10 GHz ?
2. How closely do CST port-eigenmode cutoff frequencies agree with closed-form rectangular-waveguide theory?
3. How does a square cross section change the meaning of same-index and cross-index modal S-parameters?
4. Do the electric-field, magnetic-field, and wall-current distributions support the predicted propagating and near-cutoff behavior?
5. What limitations must be considered before using this idealized section as a practical horn feed?

The evidence is analytical and simulated. The model uses vacuum filling and ideal PEC walls; finite conductor loss, fabrication tolerances, transitions, and measured validation are outside the present scope.

Contents

Portfolio Context and Evidence	i
Abstract	ii
Project Objective	iii
List of Figures	vii
List of Tables	ix
1 Hollow Waveguide Modal Analysis	1
1.1 Introduction	1
1.2 Design Definition and Analytical Baseline	1
1.2.1 Geometry and Operating Conditions	2
1.2.2 Why the Square Cross Section Matters	2
1.2.3 Expected Behavior at 10 GHz	3
1.3 Rectangular Waveguide Theory	3
1.3.1 Lowest-Order Cutoff Frequencies	3
1.3.2 Guide Wavelength and Electrical Length	4
1.3.3 TE Wave Impedance	5
1.3.4 Degenerate Modal Subspaces	5
1.4 CST Model and Simulation Method	5

1.4.1	Geometry, Materials, and Boundaries	5
1.4.2	Ports and Frequency-Domain Solution	6
1.4.3	Multimode S-Parameter Notation	6
1.4.4	Available Reproducibility Evidence	6
1.5	Simulation Results	7
1.5.1	Geometry and Port Configuration	7
1.5.2	Port-Eigenmode Cutoff Frequencies	8
1.5.3	Modal Wave Impedance	9
1.5.4	Multimode S-Parameters	10
1.5.5	Electric-Field Distributions at 10 GHz	12
1.5.6	Magnetic Field and Surface Current	14
1.6	Validation and Numerical Reliability	17
1.6.1	Analytical-to-CST Agreement	18
1.6.2	Internal Physical Consistency	18
1.6.3	Numerical and Evidence Limitations	18
1.7	Engineering Discussion	19
1.7.1	What the Degenerate Transmission Result Means	19
1.7.2	Consequences for Horn-Feed Design	19
1.7.3	Near-Cutoff Field Interpretation	20
1.7.4	Industry-Relevant Lessons	20
1.8	Engineering Evidence Matrix	20
1.9	Conclusion	24
A	Python Verification	25
A.1	Verification Script	25
A.2	Captured Console Output	27

References

29

List of Figures

1.1	CST geometry of the 20 mm $\times$ 20 mm $\times$ 90 mm hollow guide with waveguide ports at both ends.	7
1.2	Waveguide-port definitions with three solved modes per port.	8
1.3	CST cutoff-frequency markers for the first three Port 1 eigenmodes. . .	8
1.4	CST cutoff-frequency markers for the first three Port 2 eigenmodes. . .	9
1.5	CST wave-impedance results for the first three port eigenmodes.	9
1.6	Full multimode CST S-parameter response.	10
1.7	Same-mode input reflection $S_{1(1),1(1)}$	10
1.8	Same-index transmission projection $S_{2(1),1(1)}$	11
1.9	Dominant transmission into the alternate member of the degenerate output basis, $S_{2(2),1(1)}$	11
1.10	Coupling from Port 1 Mode 1 into Port 2 Mode 3, $S_{2(3),1(1)}$	12
1.11	Electric-field magnitude at 10 GHz, Port 1 Mode 1 excitation.	13
1.12	Electric-field magnitude at 10 GHz, Port 1 Mode 2 excitation.	13
1.13	Electric-field magnitude at 10 GHz, Port 1 Mode 3 excitation.	14
1.14	Magnetic-field magnitude at 10 GHz, Port 1 Mode 1 excitation.	15
1.15	Magnetic-field magnitude at 10 GHz, Port 1 Mode 2 excitation.	15
1.16	Magnetic-field magnitude at 10 GHz, Port 1 Mode 3 excitation, displayed at 270° phase.	16
1.17	PEC wall-current magnitude at 10 GHz, Port 1 Mode 1 excitation. . .	16
1.18	PEC wall-current magnitude at 10 GHz, Port 1 Mode 2 excitation. . .	17

1.19 PEC wall-current magnitude at 10 GHz, Port 1 Mode 3 excitation, displayed at 270° phase.	17
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List of Tables

1.1	Public model definition.	2
1.2	Analytical low-order modal behavior at 10 GHz.	4
1.3	Analytical and CST cutoff-frequency comparison.	9
1.4	Interpretation of selected multimode terms.	12
1.5	Engineering evidence matrix for the hollow square-waveguide study. . .	21

Chapter 1

Hollow Waveguide Modal Analysis

1.1 Introduction

Hollow metallic waveguides are used when low-loss, high-power microwave transmission and controlled modal fields are required. A rectangular guide supports transverse-electric (TE) and transverse-magnetic (TM) modes above geometry-dependent cutoff frequencies, but it cannot support a true TEM mode [1]. A horn antenna can be interpreted as a waveguide whose cross section gradually expands into a radiating aperture, so the guided field delivered by the feed section strongly influences aperture polarization and phase distribution [2].

This project uses a square special case of the rectangular waveguide. The equal transverse dimensions create a pair of degenerate lowest-order modes: TE_{10} and TE_{01} . Their identical cutoff frequencies make the modal polarization basis non-unique [1]. That feature is useful for studying port-mode ordering and multimode S-parameter interpretation, but it is usually undesirable when a practical horn feed requires a unique, polarization-controlled dominant mode.

The study combines closed-form theory, Python calculations, and a frequency-domain CST model. The main contribution is not merely reproduction of cutoff values; it is the interpretation of why nearly all transmitted power may appear in an alternate degenerate mode index at the output port even though the guide is uniform and contains no physical mode-converting discontinuity.

1.2 Design Definition and Analytical Baseline

1.2.1 Geometry and Operating Conditions

The modeled vacuum region has the dimensions

$$a = 20 \text{ mm}, \quad b = 20 \text{ mm}, \quad l = 90 \text{ mm}.$$

The transverse dimensions are equal, so the structure is a square waveguide within the broader rectangular-waveguide family. The frequency-domain simulation covers

$$8 \text{ GHz} \leq f \leq 15 \text{ GHz},$$

and the principal field-evaluation frequency is

$$f_0 = 10 \text{ GHz}.$$

Table 1.1: Public model definition.

Quantity	Value	Model interpretation
Width a	20 mm	First transverse dimension
Height b	20 mm	Second transverse dimension
Length l	90 mm	Uniform propagation section
Interior	Vacuum	Lossless filling medium
Walls	PEC	Ideal conductor, zero conductor loss
Frequency sweep	8–15 GHz	Frequency-domain solution
Field reference	10 GHz	Modal field comparison
Port modes	3 per port	Lowest three solved eigenmodes

1.2.2 Why the Square Cross Section Matters

For a conventional rectangular horn feed, one normally selects $a > b$ so that the TE_{10} mode has a lower cutoff frequency than TE_{01} [1], [2]. Here, $a = b$, which forces

$$f_{c,10} = f_{c,01}.$$

The model therefore supports two orthogonally polarized lowest-order modes at the same cutoff frequency. A numerical port solver may choose any orthogonal basis spanning that degenerate subspace. The selected basis can differ between Port 1 and Port 2, so mode index alone is not a reliable physical polarization label [1].

1.2.3 Expected Behavior at 10 GHz

Using the exact vacuum speed of light, the first two cutoff frequencies are expected near 7.4948 GHz. The (1,1)-family cutoff is expected near 10.5993 GHz. Therefore:

- the first two solved modes should propagate at 10 GHz;
- the third solved mode should be slightly below cutoff at 10 GHz;
- the first two port modes should exhibit nearly identical propagation constants and wave impedances;
- the output modal amplitudes may be rotated between the two degenerate basis functions.

1.3 Rectangular Waveguide Theory

For a vacuum-filled rectangular waveguide, the cutoff frequency of a TE or TM mode is [1]

$$f_{c,mn} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}.$$

The transverse cutoff wavenumber and longitudinal propagation constant are

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}, \quad \beta_{mn} = \sqrt{k_0^2 - k_c^2},$$

where $k_0 = 2\pi f/c$. A mode is propagating when $f > f_c$ and evanescent when $f < f_c$.

1.3.1 Lowest-Order Cutoff Frequencies

For $a = b = 20$ mm,

$$f_{c,10} = f_{c,01} = \frac{c}{2a} = 7.49481145 \text{ GHz}.$$

The next cutoff family is

$$f_{c,11} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = 10.599264 \text{ GHz.}$$

The TE₂₀ and TE₀₂ modes begin near

$$f_{c,20} = f_{c,02} = \frac{c}{a} = 14.989623 \text{ GHz.}$$

Table 1.2: Analytical low-order modal behavior at 10 GHz.

Mode family	Cutoff	Status at 10 GHz	Comment
TE ₁₀	7.4948 GHz	Propagating	Degenerate lowest-order pair
TE ₀₁	7.4948 GHz	Propagating	Orthogonal polarization partner
(1, 1) TE/TM family	10.5993 GHz	Evanescant	Slightly below cutoff
TE ₂₀ /TE ₀₂	14.9896 GHz	Evanescant	Near upper sweep boundary

1.3.2 Guide Wavelength and Electrical Length

For a propagating mode,

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - (f_c/f)^2}}, \quad \lambda_0 = \frac{c}{f}.$$

At 10 GHz,

$$\lambda_0 = 29.979 \text{ mm,}$$

and for either member of the lowest-order pair,

$$\lambda_g = 45.284 \text{ mm.}$$

The 90 mm physical section is therefore

$$\frac{l}{\lambda_g} = 1.987$$

guide wavelengths long.

1.3.3 TE Wave Impedance

The TE-mode wave impedance is

$$Z_{\text{TE}} = \frac{\eta_0}{\sqrt{1 - (f_c/f)^2}}.$$

Using $\eta_0 = 376.730 \, \Omega$, the lowest-order TE impedance at 10 GHz is

$$Z_{\text{TE}} \approx 569.06 \, \Omega.$$

This modal impedance is not a fixed $50 \, \Omega$ normalization. A waveguide port solves the transverse eigenfield and computes its frequency-dependent propagation and impedance characteristics [3], [4].

1.3.4 Degenerate Modal Subspaces

When two modes have the same eigenvalue, any orthogonal linear combination of those modes is also a valid eigenbasis [1]. For the square guide, a solver may return one pair of orthogonal transverse field patterns at the input and a rotated pair at the output. If $\mathbf{u}_1, \mathbf{u}_2$ span the input degenerate subspace and $\mathbf{v}_1, \mathbf{v}_2$ span the output subspace, then

$$\begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \end{bmatrix} = \mathbf{R} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{bmatrix},$$

where $\mathbf{R}$ is an orthogonal basis rotation. Consequently, a near-zero $S_{2(1),1(1)}$ and near-unity $S_{2(2),1(1)}$ can represent the same physical polarization transported through a uniform guide, merely expressed in a different numerical basis.

1.4 CST Model and Simulation Method

1.4.1 Geometry, Materials, and Boundaries

The CST model represents the hollow interior as a vacuum brick. The surrounding background is PEC, so the brick boundary acts as an ideal conducting wall. Tangential electric field therefore satisfies

$$\mathbf{E}_t = 0 \quad \text{on the PEC boundary.}$$

No finite wall thickness or conductivity is modeled. The simulation therefore isolates modal geometry from conductor loss.

1.4.2 Ports and Frequency-Domain Solution

Waveguide ports are assigned at the two longitudinal end faces. Port 1 is located at $z = 0$ and points in the $+z$ direction. Port 2 is located at $z = 90$ mm and points in the $-z$ direction. Each port solves three eigenmodes. This workflow follows the official CST Learning Edition hollow-waveguide example and the documented use of waveguide ports for modal excitation [3], [4].

The frequency-domain solver is used over 8–15 GHz. The public source contains figures at 10 GHz for electric field, magnetic field, and wall surface current associated with the first three Port 1 modal excitations.

1.4.3 Multimode S-Parameter Notation

Because there are two ports and three solved modes per port, the result is a multimode scattering matrix. The notation

$$S_{q(j),p(i)}$$

denotes the complex wave amplitude observed at Port q , Mode j , when Port p , Mode i is excited. Relevant examples are:

- $S_{1(1),1(1)}$: same-mode input reflection;
- $S_{2(1),1(1)}$: projection onto output Mode 1;
- $S_{2(2),1(1)}$: projection onto output Mode 2;
- $S_{2(3),1(1)}$: coupling into the third solved output mode.

For a degenerate pair, the physically meaningful transmitted power is associated with the total power in the degenerate subspace rather than one same-index coefficient.

1.4.4 Available Reproducibility Evidence

The native CST project is not part of the public Git history. Reproducibility is therefore limited to:

- the documented geometry, materials, frequency range, and port configuration;
- the analytical equations and Python verification;
- the selected CST plots and numerical cutoff markers;
- explicit notes on missing convergence and model artifacts.

1.5 Simulation Results

1.5.1 Geometry and Port Configuration

Figure 1.1 shows the final vacuum region with waveguide ports on both longitudinal end faces. Figures 1.2a and 1.2b document the opposite port orientations and the three-mode setting.

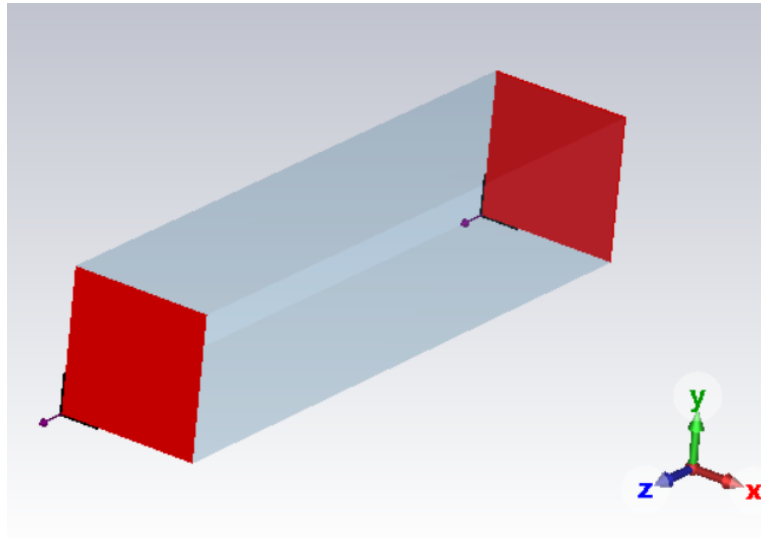
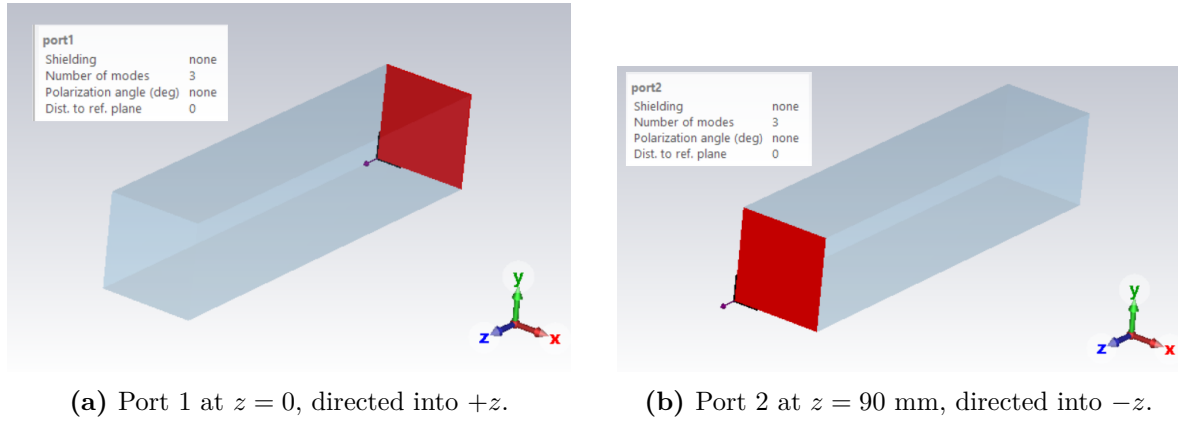


Figure 1.1: CST geometry of the 20 mm $\times$ 20 mm $\times$ 90 mm hollow guide with waveguide ports at both ends.

(a) Port 1 at $z = 0$, directed into $+z$.(b) Port 2 at $z = 90$ mm, directed into $-z$.**Figure 1.2:** Waveguide-port definitions with three solved modes per port.

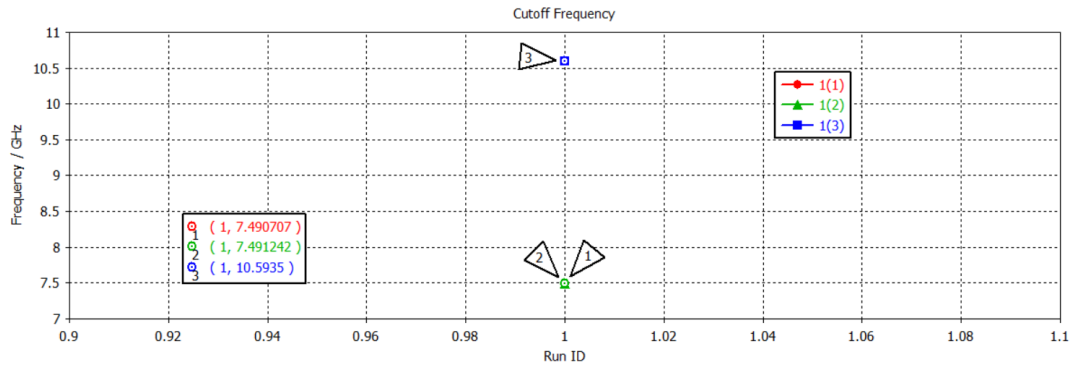
1.5.2 Port-Eigenmode Cutoff Frequencies

CST reports the following cutoff frequencies:

Port 1: 7.490707, 7.491242, 10.5935 GHz,

Port 2: 7.49031, 7.49064, 10.5931 GHz.

Figures 1.3 and 1.4 show the marked CST values.

**Figure 1.3:** CST cutoff-frequency markers for the first three Port 1 eigenmodes.

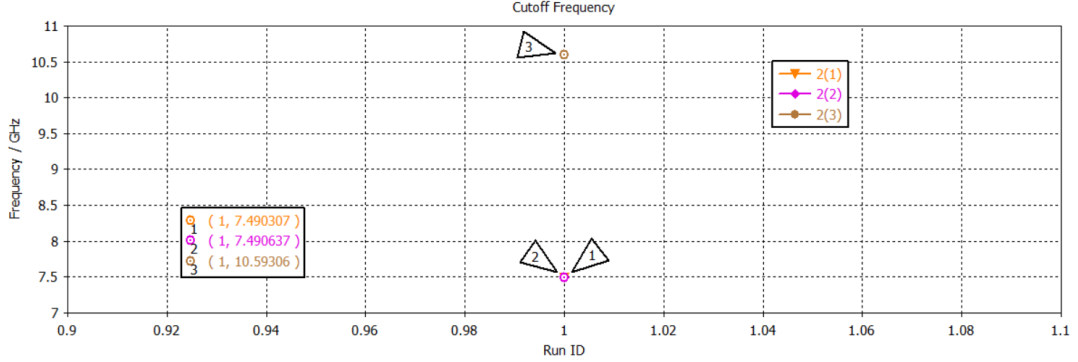


Figure 1.4: CST cutoff-frequency markers for the first three Port 2 eigenmodes.

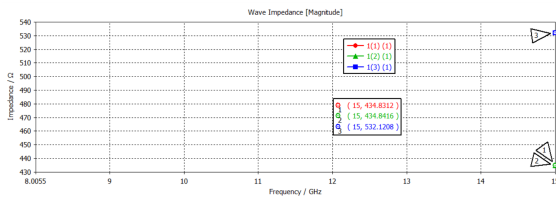
Table 1.3: Analytical and CST cutoff-frequency comparison.

Solved mode	Theory	Port 1	Port 1 error	Port 2	Port 2 error
Mode 1	7.494811 GHz	7.490707 GHz	−0.0548%	7.490310 GHz	−0.0601%
Mode 2	7.494811 GHz	7.491242 GHz	−0.0476%	7.490640 GHz	−0.0557%
Mode 3	10.599264 GHz	10.593500 GHz	−0.0544%	10.593100 GHz	−0.0582%

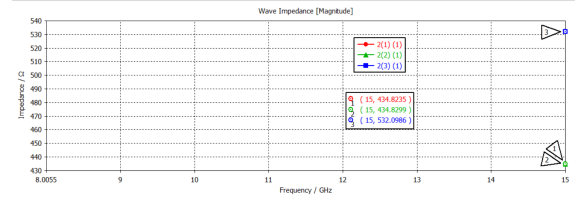
All six differences remain below 0.061%. The agreement supports the geometry definition and port-eigenmode solution.

1.5.3 Modal Wave Impedance

Figures 1.5a and 1.5b show the frequency-dependent modal wave impedances. The first two traces are expected to overlap closely because they belong to the degenerate lowest-order TE pair.



(a) Port 1 modal wave impedance.



(b) Port 2 modal wave impedance.

Figure 1.5: CST wave-impedance results for the first three port eigenmodes.

The dominant analytical value at 10 GHz is 569.06 Ω . The plots are consistent with a high TE impedance near cutoff and confirm that the ports are normalized to modal wave impedance rather than a fixed coaxial value.

1.5.4 Multimode S-Parameters

Figure 1.6 shows the complete two-port, three-mode scattering response.

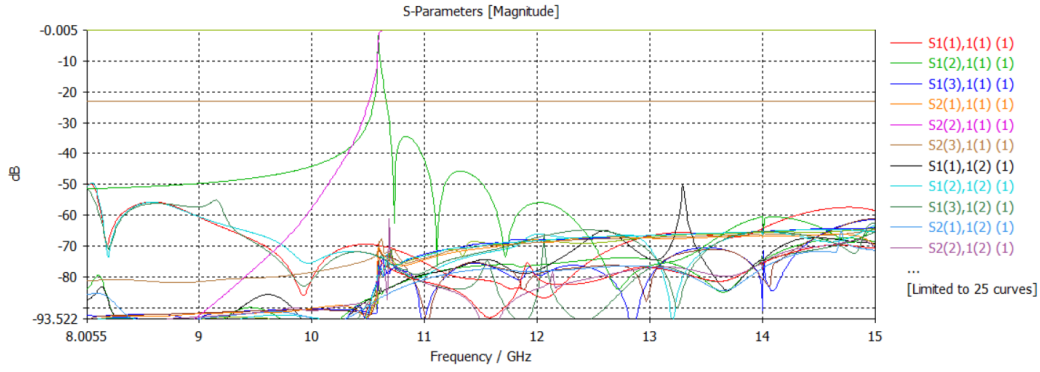


Figure 1.6: Full multimode CST S-parameter response.

The same-mode reflection coefficient $S_{1(1),1(1)}$ remains below approximately -50 dB across the plotted 8–15 GHz band and is near -85 dB around 10 GHz by visual reading of the trace. These values are plot-derived estimates because the public package does not contain exported numerical S-parameter data.

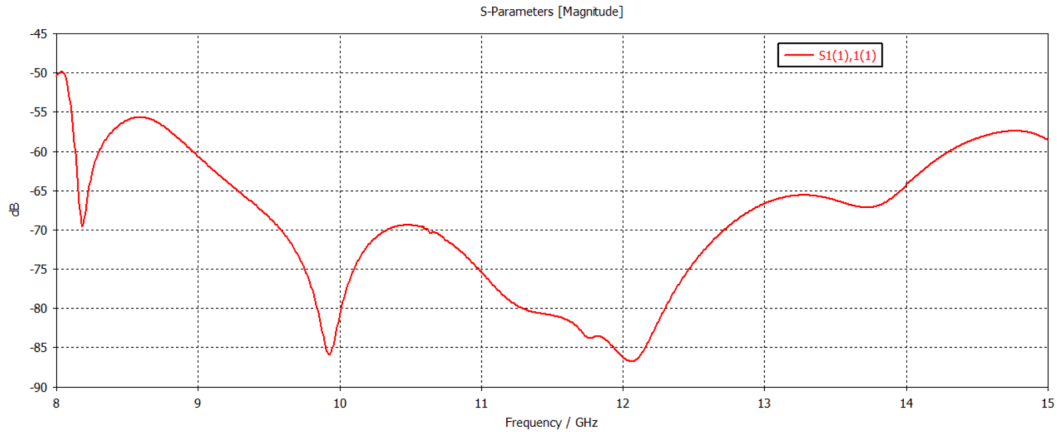


Figure 1.7: Same-mode input reflection $S_{1(1),1(1)}$.

The same-index transmission term $S_{2(1),1(1)}$ does not capture the total transmitted field because the first two modes are degenerate.

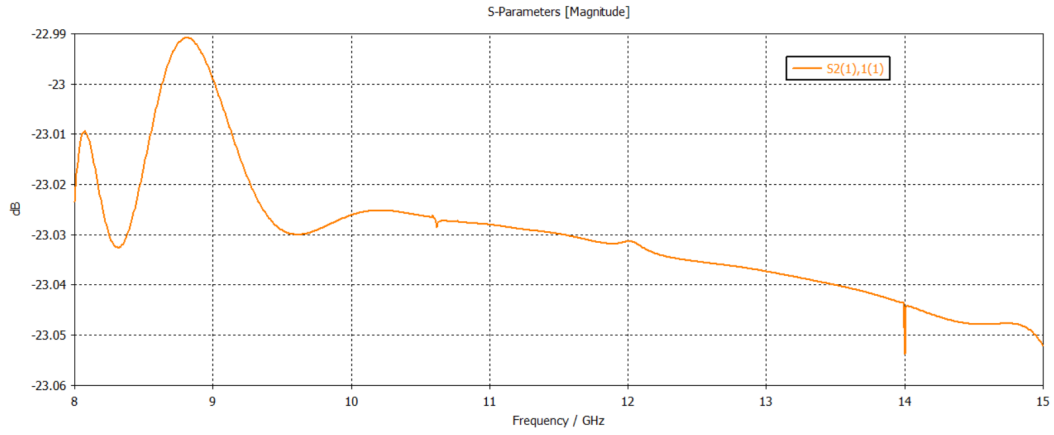


Figure 1.8: Same-index transmission projection $S_{2(1),1(1)}$.

The alternate-basis coefficient $S_{2(2),1(1)}$ stays close to 0 dB, approximately -0.02 dB near 10 GHz by visual reading. This is the dominant output projection.

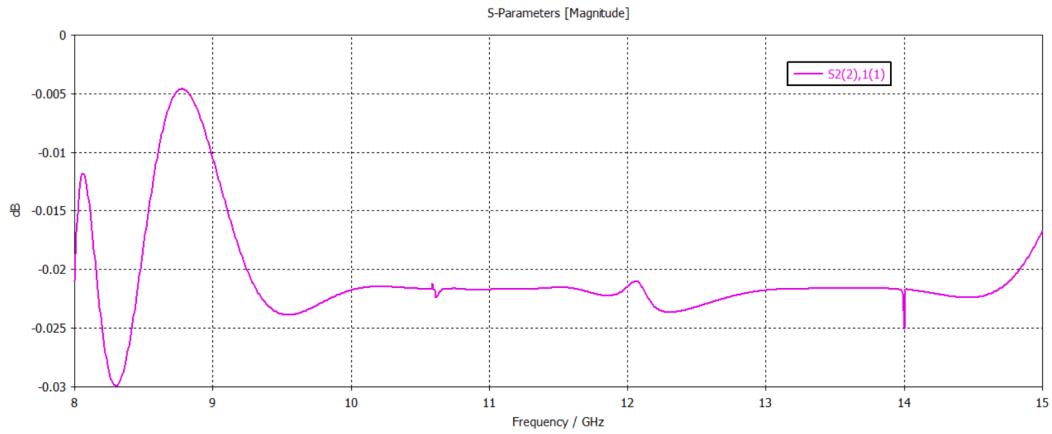


Figure 1.9: Dominant transmission into the alternate member of the degenerate output basis, $S_{2(2),1(1)}$.

Coupling to the third solved output mode is much smaller.

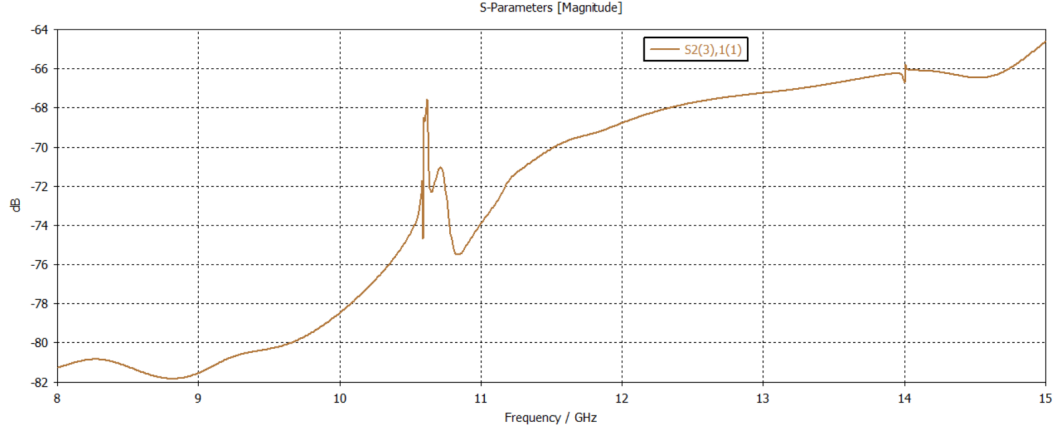


Figure 1.10: Coupling from Port 1 Mode 1 into Port 2 Mode 3, $S_{2(3),1(1)}$.

Table 1.4: Interpretation of selected multimode terms.

CST term	Meaning	Interpretation
$S_{1(1),1(1)}$	Mode-1 input reflection	Very small in the ideal uniform guide
$S_{2(1),1(1)}$	Output Mode-1 projection	Not total transmission in a degenerate subspace
$S_{2(2),1(1)}$	Output Mode-2 projection	Dominant near-unity transmitted coefficient
$S_{2(3),1(1)}$	Third-mode projection	Weak higher-order coupling

1.5.5 Electric-Field Distributions at 10 GHz

Figures 1.11–1.13 show the electric-field magnitude for excitation from the first three Port 1 modes.

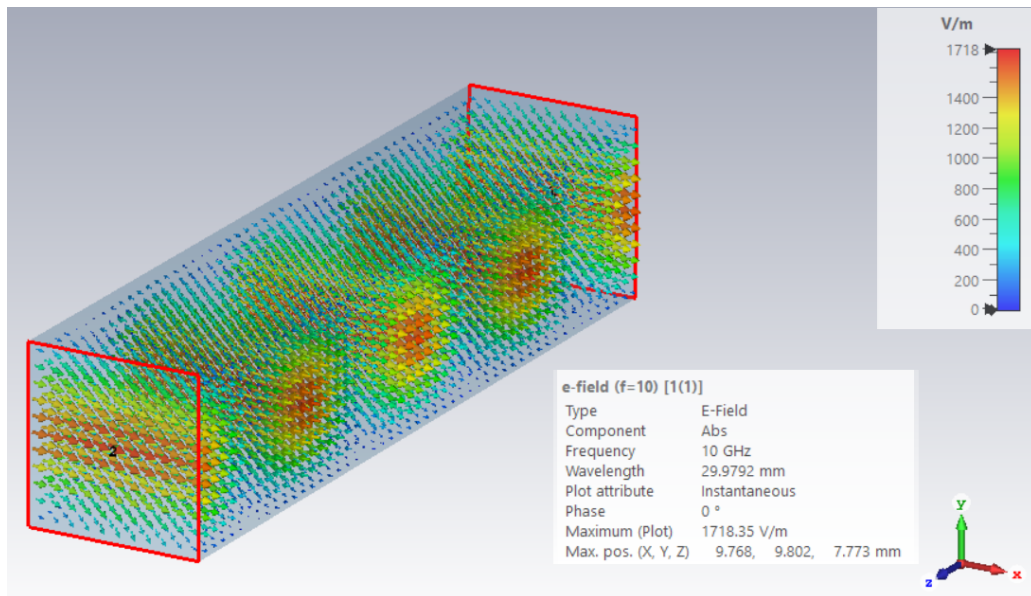


Figure 1.11: Electric-field magnitude at 10 GHz, Port 1 Mode 1 excitation.

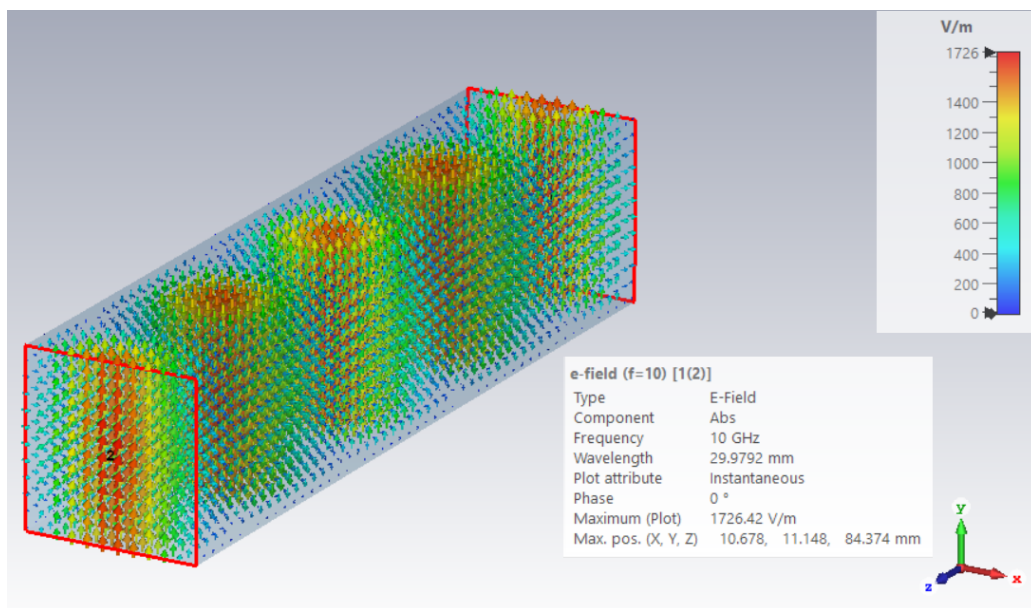


Figure 1.12: Electric-field magnitude at 10 GHz, Port 1 Mode 2 excitation.

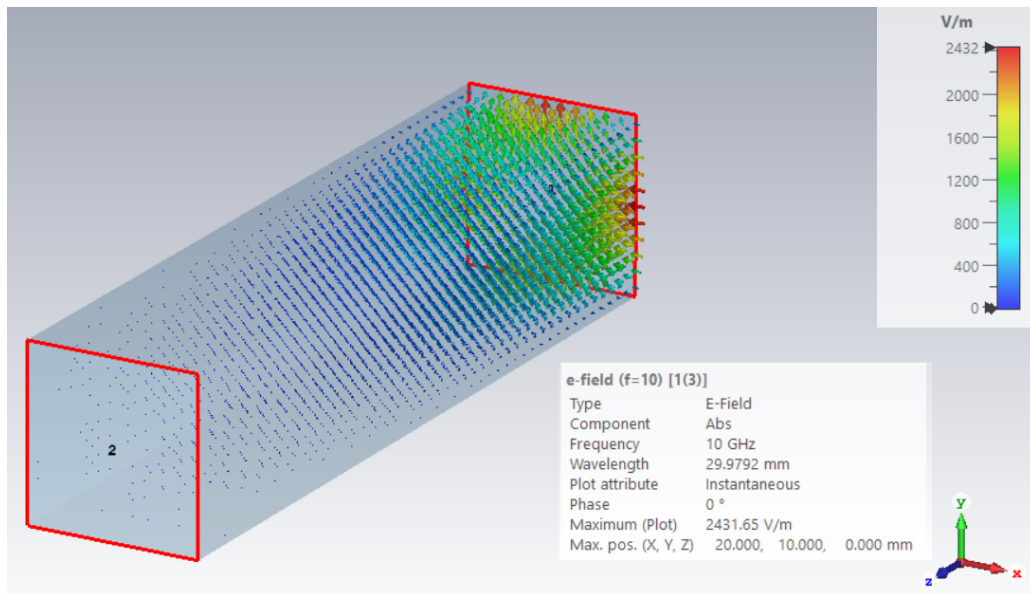


Figure 1.13: Electric-field magnitude at 10 GHz, Port 1 Mode 3 excitation.

The first two fields extend throughout the guide, consistent with propagation above cutoff. The third field is more localized because 10 GHz is below its 10.5935 GHz CST cutoff.

1.5.6 Magnetic Field and Surface Current

The magnetic-field results are shown in Figures 1.14–1.16. The third-mode view uses a 270° phase reference for visibility.

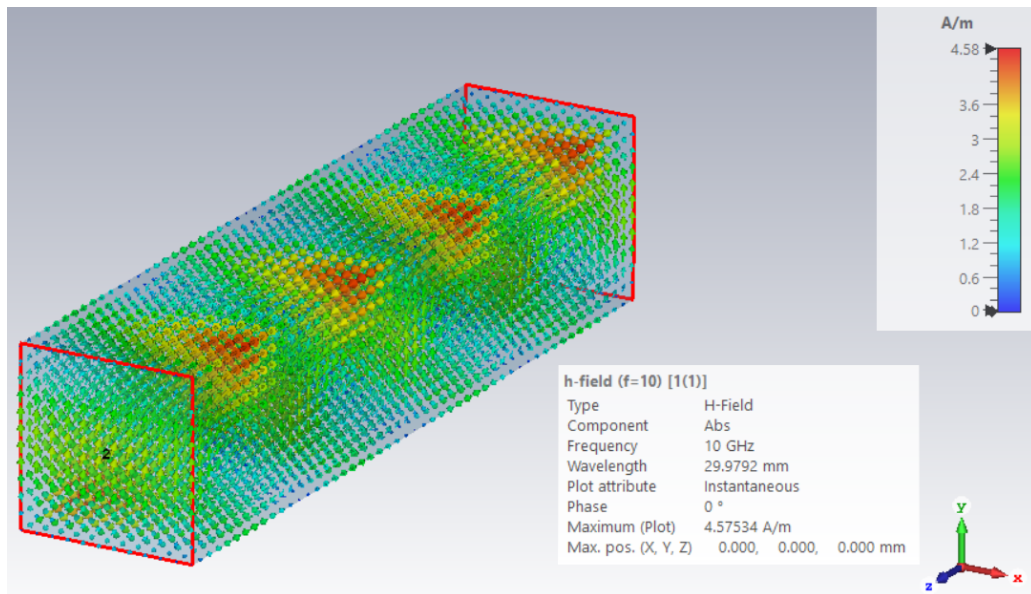


Figure 1.14: Magnetic-field magnitude at 10 GHz, Port 1 Mode 1 excitation.

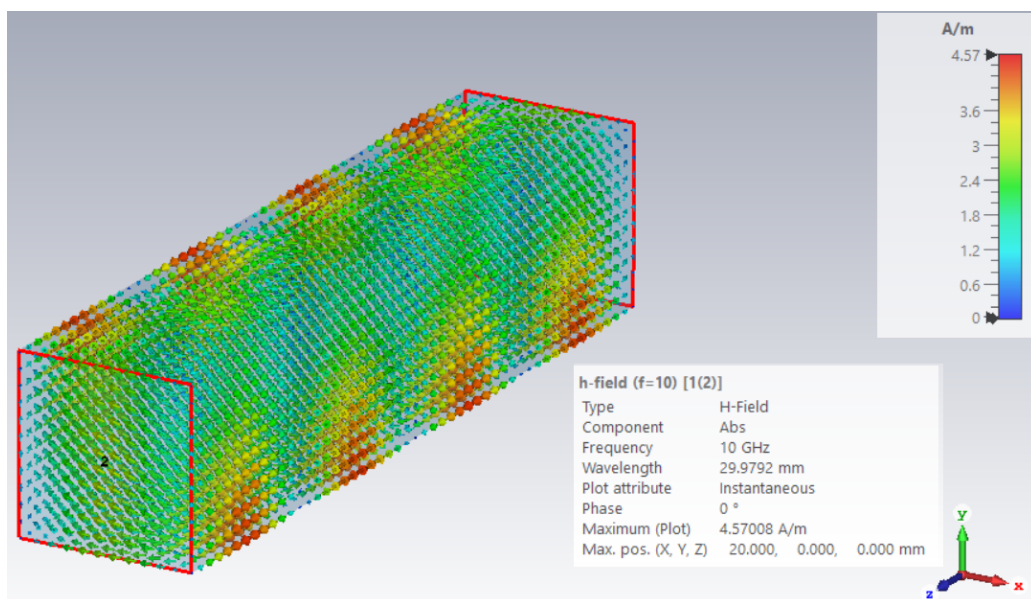


Figure 1.15: Magnetic-field magnitude at 10 GHz, Port 1 Mode 2 excitation.

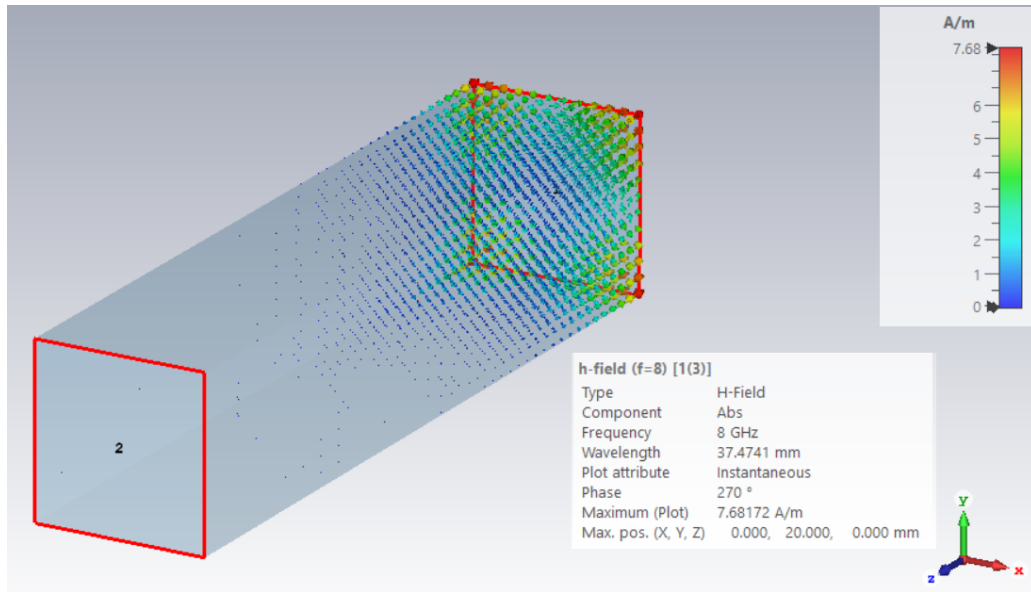


Figure 1.16: Magnetic-field magnitude at 10 GHz, Port 1 Mode 3 excitation, displayed at 270° phase.

Figures 1.17–1.19 show the corresponding wall-current distributions induced by the modal fields.

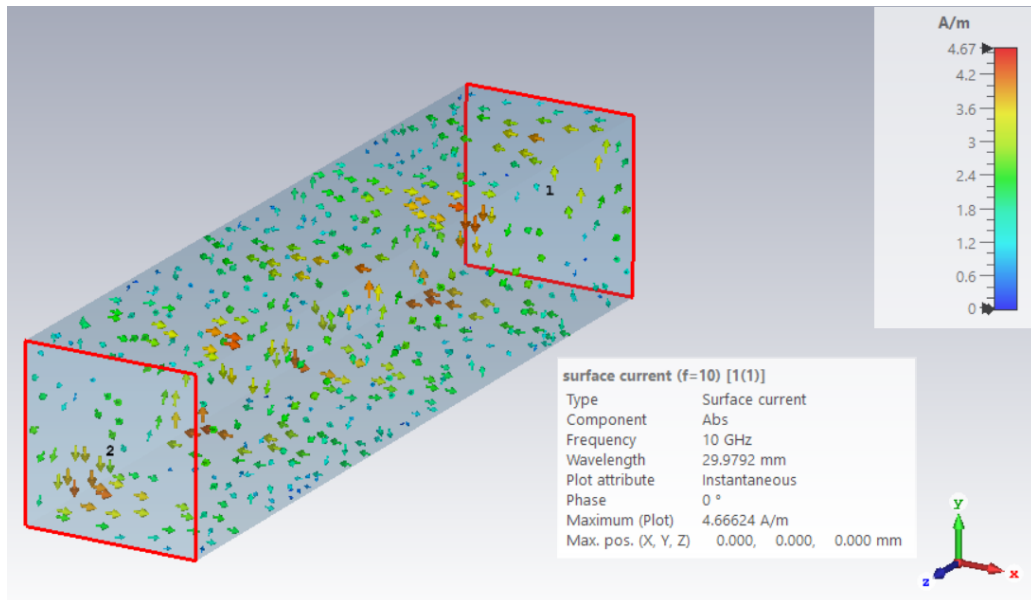


Figure 1.17: PEC wall-current magnitude at 10 GHz, Port 1 Mode 1 excitation.

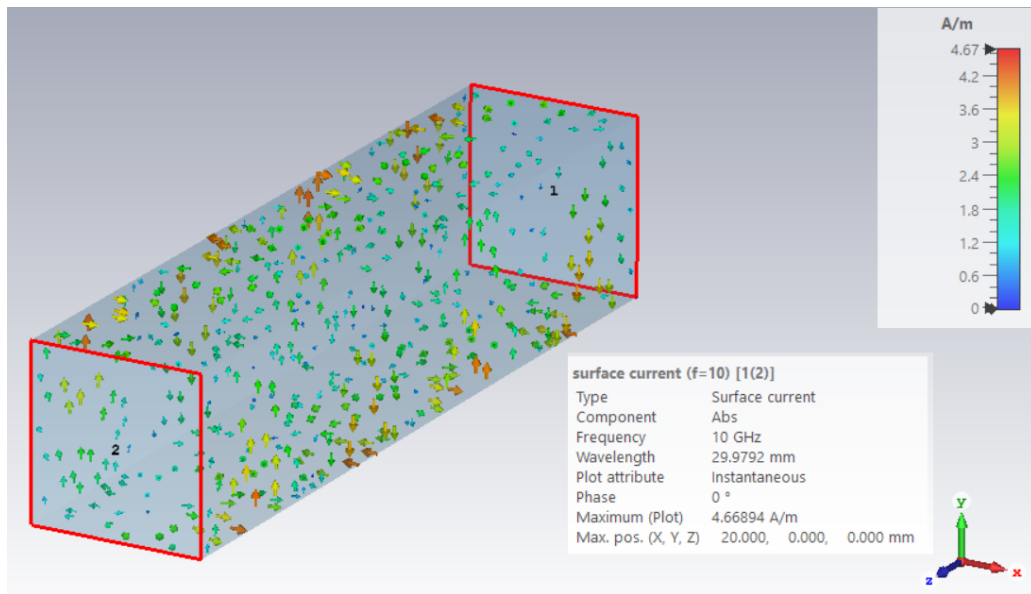


Figure 1.18: PEC wall-current magnitude at 10 GHz, Port 1 Mode 2 excitation.

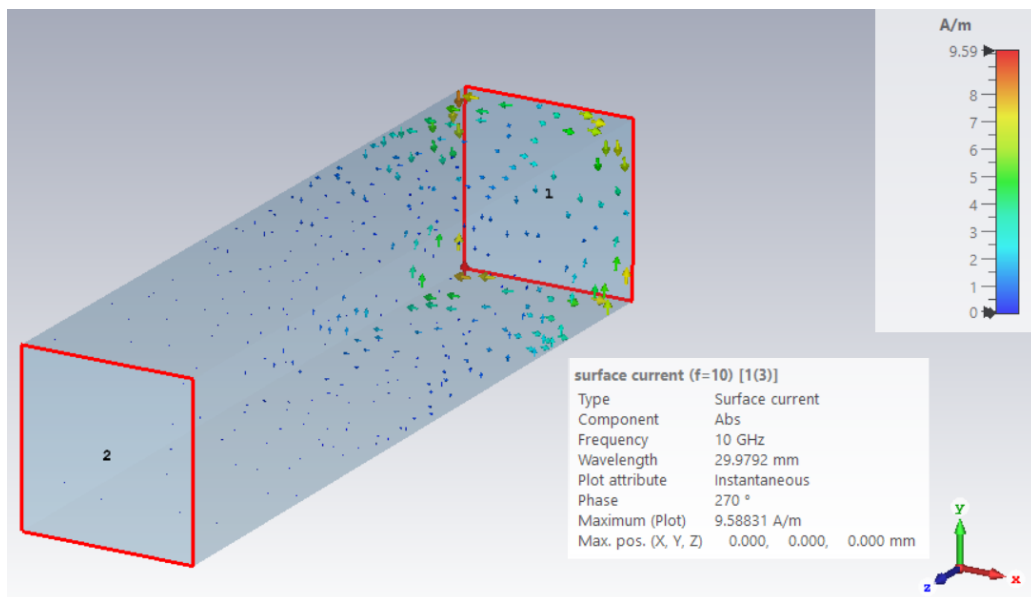


Figure 1.19: PEC wall-current magnitude at 10 GHz, Port 1 Mode 3 excitation, displayed at 270° phase.

1.6 Validation and Numerical Reliability

1.6.1 Analytical-to-CST Agreement

The strongest quantitative validation is the cutoff-frequency comparison in Table 1.3. Every CST value differs from the corresponding closed-form result by less than 0.061%. The two ports also agree closely with one another, as expected for a uniform symmetric structure.

1.6.2 Internal Physical Consistency

The available evidence is mutually consistent:

- two nearly equal port cutoffs occur near 7.49 GHz;
- both corresponding fields propagate at 10 GHz;
- the third cutoff is near 10.59 GHz;
- the third field is localized at 10 GHz;
- same-index transmission is not dominant, while the alternate degenerate-basis coefficient is near unity;
- weak projection into the third solved mode is consistent with a uniform guide and orthogonality.

1.6.3 Numerical and Evidence Limitations

The following limitations constrain the conclusions:

1. **No formal mesh convergence.** The public record does not include a mesh-density sweep or adaptive-convergence history.
2. **No port-basis convergence study.** Polarization lines, basis orientation, and the number of solved modes were not varied systematically.
3. **Ideal materials.** Vacuum filling and PEC walls eliminate conductor and dielectric loss.
4. **No raw data exports.** S-parameter magnitudes quoted away from cutoff markers are approximate visual readings from figures.
5. **No native CST model.** Geometry and settings are documented, but the binary model is not distributed through ordinary Git history.

6. **Only three solved modes shown.** The square guide has additional degenerate mode structure near the (1, 1) cutoff; the third displayed mode should therefore be described as a (1, 1)-family solution rather than assigned an unverified unique TE/TM identity.
7. **No horn flare or transition.** The study analyzes a uniform guide, not a complete radiating horn or feed transition.
8. **No measurement.** No prototype, VNA calibration, field probe, or chamber result is included.

1.7 Engineering Discussion

1.7.1 What the Degenerate Transmission Result Means

The most important result is the apparent transfer from Port 1 Mode 1 primarily into Port 2 Mode 2. A superficial reading could suggest strong physical conversion between orthogonal modes. That interpretation would be misleading because the guide is uniform and the first two modes are exactly degenerate in the ideal geometry.

The correct interpretation is basis rotation. The input and output port eigensolvers can assign different orthogonal vectors to the same two-dimensional degenerate subspace. The coefficient $S_{2(2),1(1)}$ then becomes large while $S_{2(1),1(1)}$ becomes small. Total transmitted power across the two-mode subspace remains the relevant quantity.

1.7.2 Consequences for Horn-Feed Design

A square guide is useful for dual-polarization systems when polarization is intentionally controlled by transitions, probes, irises, or orthomode transducers. It is less suitable as an uncontrolled single-polarization horn feed because small asymmetries, manufacturing tolerances, bends, and transitions can select or mix the degenerate pair.

A conventional single-polarization rectangular horn normally uses $a > b$ and operates in a band satisfying

$$f_{c,10} < f < \min(f_{c,20}, f_{c,01}),$$

so that only TE₁₀ propagates [1], [2]. The present $a = b$ model does not provide that single-mode separation. Therefore, the phrase “horn-feed application” should be understood as a guided-wave modal foundation, not as a finalized practical feed specification.

1.7.3 Near-Cutoff Field Interpretation

The third solved mode has a cutoff approximately 5.9% above 10 GHz. Its longitudinal propagation constant is imaginary at the observation frequency, so the mode is evanescent. The localized electric, magnetic, and current patterns are consistent with this behavior.

Changing the displayed phasor phase can make near-cutoff field components more visible, but it does not transform an evanescent mode into a propagating one. Absolute magnitude plots and phase-referenced snapshots are visualization choices; propagation status is determined by the eigenvalue and cutoff relation.

1.7.4 Industry-Relevant Lessons

The project demonstrates several practices that extend beyond a classroom waveguide model:

- verify port-mode labels with field orientation rather than relying only on mode index;
- sum power across a degenerate modal subspace when interpreting multimode transmission;
- use cutoff calculations before selecting simulation frequency and port modes;
- separate ideal PEC modal validation from finite-conductivity insertion-loss prediction;
- document solver warnings when symmetry or degeneracy makes the numerical basis non-unique;
- avoid claiming a production horn-feed design without transition, tolerance, thermal, and measurement evidence.

1.8 Engineering Evidence Matrix

The matrix in Table 1.5 connects each principal engineering claim to its analytical basis, simulation record, quantitative outcome, evidence disposition, and remaining validation gap. This structure separates what is directly verified from what is only supported qualitatively, assumed by the ideal model, or not yet established.

Table 1.5: Engineering evidence matrix for the hollow square-waveguide study.

Engineering claim or decision	Evidence record and trace	Quantitative outcome	Disposition	Residual gap or next validation
The public model is a uniform ideal hollow square guide.	Design definition, CST geometry in Figure 1.1, and documented port locations.	Interior dimensions: $20 \times 20 \times 90$ mm; vacuum filling; PEC walls; 8–15 GHz sweep.	Verified model definition	Native CST binary is not distributed; dimensional tolerances, wall thickness, and finite conductivity are not represented.
The two lowest-order TE solutions are degenerate.	Closed-form rectangular-waveguide theory [1], Python verification, and both CST port-eigenmode solutions in Table 1.3.	Theory: 7.494811 GHz. CST: 7.490310–7.491242 GHz. Maximum theory-to-CST difference: 0.0601%.	Strong within ideal model	Repeat with controlled polarization lines and a small deliberate geometric perturbation to test basis sensitivity.
The third solved mode is below cutoff at 10 GHz.	Closed-form cutoff, CST cutoff markers, and localized electric-, magnetic-, and wall-current fields in Section 1.5.	Theory: 10.599264 GHz. CST: approximately 10.5931–10.5935 GHz.	Strong within ideal model	Solve and classify additional modes near the (1, 1)-family cutoff; export propagation constants for direct numerical confirmation.
Lowest-order propagation quantities at 10 GHz are consistent with TE theory.	Closed-form calculation and Python output; CST modal-impedance traces in Figures 1.5a and 1.5b.	$\lambda_g = 45.284$ mm, $Z_{\text{TE}} = 569.057 \, \Omega$, and guide length $= 1.987\lambda_g$.	Quantitatively supported	CST impedance values were not exported as numerical data; add marker exports or ASCII result files for point-by-point comparison.

Continued on next page

Table 1.5 – continued from previous page

Engineering claim or decision	Evidence record and trace	Quantitative outcome	Disposition	Residual gap or next validation
The uniform ideal guide has very small same-mode input reflection.	CST trace in Figure 1.7; values are read visually from the plotted response.	Below approximately -50 dB over the plotted band and near -85 dB around 10 GHz.	Qualitative plot evidence	Export Touchstone or ASCII data and report exact values, reference impedance, solver accuracy, and energy balance.
Dominant transmitted amplitude appears in an alternate output mode index because of degenerate-basis rotation.	Multimode traces in Figures 1.8 and 1.9, plus degenerate-subspace theory [1].	$S_{2(2),1(1)}$ is approximately -0.02 dB near 10 GHz, while the same-index projection is much smaller.	Strong qualitative interpretation	Compute total transmitted power summed over the degenerate subspace and repeat with varied port-basis orientation.
Field behavior is consistent with modal propagation status.	Electric-field, magnetic-field, and surface-current figures for the first three Port 1 excitations.	Modes 1 and 2 extend through the guide; Mode 3 is spatially localized at 10 GHz.	Internally consistent	Add axial field samples, decay-constant extraction, and normalized modal-overlap integrals.
Negligible insertion loss is a model consequence, not a hardware result.	PEC walls and vacuum filling specified in the CST model; no finite-conductivity or dielectric-loss calculation.	Lossless idealization by construction.	Assumption only	Replace PEC with copper, document conductivity and surface roughness, and perform mesh-converged attenuation analysis.

Continued on next page

Table 1.5 – continued from previous page

Engineering claim or decision	Evidence record and trace	Quantitative outcome	Disposition	Residual gap or next validation
The present structure is not qualified as a practical horn feed.	Single-mode design criterion from waveguide and antenna theory [1], [2]; no transition, flare, tolerance, thermal, or radiation model.	Square guide has no unique lowest-order polarization because $a = b$; complete horn performance was not simulated.	Not established	Compare with an $a > b$ guide, add a feed transition and horn flare, and evaluate match, polarization, gain, pattern, and tolerance sensitivity.
The public package is reproducible at the documentation and analytical levels.	Modular LaTeX source, Python verification script, captured console output, curated CST figures, result tables, and SHA-256 manifests.	Report compiles with <code>latexmk</code> and Biber; analytical quantities regenerate from the included script.	Reproducible public record	Full electromagnetic reproducibility still requires the native CST model, solver version, mesh history, and exported numerical results.
Experimental performance has not been validated.	No prototype, calibration record, VNA data, field probe, or chamber measurement is included.	No measured cutoff, insertion loss, phase delay, or radiation result.	Not performed	Fabricate a controlled guide section and validate with calibrated S-parameter and field measurements, including uncertainty reporting.

The matrix supports three final engineering conclusions. First, the cutoff and propagation-state claims are strongly supported within the ideal PEC-vacuum model. Second, the unusual output mode ordering is a credible degenerate-basis interpretation but still benefits from a dedicated port-basis sensitivity study. Third, practical horn-feed suitability, insertion loss, and measured performance remain open validation items rather than demonstrated outcomes.

1.9 Conclusion

An ideal $20\text{ mm} \times 20\text{ mm} \times 90\text{ mm}$ hollow square waveguide was analyzed at 10 GHz using closed-form theory, Python verification, and CST Studio Suite.

Theory predicts two degenerate lowest-order cutoff frequencies of 7.494811 GHz and a $(1, 1)$ -family cutoff of 10.599264 GHz. CST reports nearly identical values at both ports, with all relative differences below 0.061%. The first two modes therefore propagate at 10 GHz, while the third solved mode remains slightly below cutoff.

The multimode S-parameter response illustrates a central consequence of square-guide degeneracy: most transmitted amplitude appears in the alternate output mode index. This is best understood as a rotation of the numerical eigenbasis within the degenerate subspace rather than physical conversion caused by a discontinuity. Electric-field, magnetic-field, and wall-current plots are consistent with propagating behavior for the first two modes and localized near-cutoff behavior for the third.

The project provides a strong modal-analysis and simulation-interpretation case study. It does not establish a practical horn-feed design because the public record lacks finite conductivity, port-basis sensitivity, mesh convergence, transitions, a horn flare, fabrication tolerances, and measured validation. Future work should intentionally break or control the degeneracy, compare a conventional $a > b$ guide, include copper loss, model a horn transition, and validate the design with exported data and measurement.

Appendix A

Python Verification

The following script reproduces the analytical cutoff frequencies, propagation status, guide wavelength, TE wave impedance, electrical length, and CST cutoff-error comparison used in the public report.

A.1 Verification Script

```
1  """Verify the analytical modal quantities for the public waveguide project."""
2
3  from __future__ import annotations
4
5  import math
6  from dataclasses import dataclass
7
8  C0 = 299_792_458.0
9  ETA0 = 376.730313668
10 F0 = 10.0e9
11
12 A = 20.0e-3
13 B = 20.0e-3
14 LENGTH = 90.0e-3
15
16 CST_PORT1_GHZ = (7.490707, 7.491242, 10.5935)
17 CST_PORT2_GHZ = (7.49031, 7.49064, 10.5931)
18
19
20 @dataclass(frozen=True)
21 class Mode:
22     name: str
23     m: int
24     n: int
25
```

```

26
27 MODES = (
28     Mode("TE10", 1, 0),
29     Mode("TE01", 0, 1),
30     Mode("(1,1) family", 1, 1),
31     Mode("TE20", 2, 0),
32     Mode("TE02", 0, 2),
33 )
34
35
36 def cutoff_frequency(mode: Mode) -> float:
37     return (C0 / 2.0) * math.sqrt((mode.m / A) ** 2 + (mode.n / B) ** 2)
38
39
40 def guide_wavelength(frequency: float, cutoff: float) -> float | None:
41     if frequency <= cutoff:
42         return None
43     lambda0 = C0 / frequency
44     return lambda0 / math.sqrt(1.0 - (cutoff / frequency) ** 2)
45
46
47 def te_impedance(frequency: float, cutoff: float) -> float | None:
48     if frequency <= cutoff:
49         return None
50     return ETA0 / math.sqrt(1.0 - (cutoff / frequency) ** 2)
51
52
53 def relative_error_percent(numerical: float, analytical: float) -> float:
54     return 100.0 * (numerical - analytical) / analytical
55
56
57 def main() -> None:
58     print("Hollow Square Waveguide Analytical Verification")
59     print("=" * 88)
60     print(f"Geometry: a = {A * 1e3:.3f} mm, b = {B * 1e3:.3f} mm, "
61           f"length = {LENGTH * 1e3:.3f} mm")
62     print(f"Observation frequency: {F0 / 1e9:.3f} GHz")
63     print()
64
65     print("Low-order analytical modes")
66     print("-" * 88)
67     print(f"{'Mode':<18}{'fc (GHz)':>12}{'Status at 10 GHz':>24}"
68           f"{'lambda_g (mm)':>18}{'Z_TE (ohm)':>16}")
69     print("-" * 88)
70
71     for mode in MODES:
72         fc = cutoff_frequency(mode)
73         lambda_g = guide_wavelength(F0, fc)
74         z_te = te_impedance(F0, fc)
75         status = "propagating" if F0 > fc else "evanescent"
76         lambda_text = f"{lambda_g * 1e3:.3f}" if lambda_g else "N/A"
77         impedance_text = f"{z_te:.3f}" if z_te else "N/A"

```

```

78     print(
79         f"{mode.name:<18}{fc / 1e9:>12.6f}{status:>24}"
80         f"{lambda_text:>18}{impedance_text:>16}"
81     )
82
83     fc10 = cutoff_frequency(MODES[0])
84     lambda_g10 = guide_wavelength(F0, fc10)
85     z_te10 = te_impedance(F0, fc10)
86     assert lambda_g10 is not None
87     assert z_te10 is not None
88
89     print()
90     print("Lowest-order detailed quantities")
91     print("-" * 88)
92     print(f"Free-space wavelength: {C0 / F0 * 1e3:.6f} mm")
93     print(f"Guide wavelength: {lambda_g10 * 1e3:.6f} mm")
94     print(f"TE wave impedance: {z_te10:.6f} ohm")
95     print(f"Electrical length: {LENGTH / lambda_g10:.6f} guide wavelengths")
96
97     analytical_ghz = (
98         cutoff_frequency(MODES[0]) / 1e9,
99         cutoff_frequency(MODES[1]) / 1e9,
100        cutoff_frequency(MODES[2]) / 1e9,
101    )
102
103    print()
104    print("CST cutoff comparison")
105    print("-" * 88)
106    print(f"{'Mode':<10}{'Theory':>14}{'Port 1':>14}{'Error P1':>14}"
107          f"{'Port 2':>14}{'Error P2':>14}")
108    print("-" * 88)
109
110    for index, theory in enumerate(analytical_ghz, start=1):
111        p1 = CST_PORT1_GHZ[index - 1]
112        p2 = CST_PORT2_GHZ[index - 1]
113        print(
114            f"{index:<10}{theory:>14.6f}{p1:>14.6f}"
115            f"{relative_error_percent(p1, theory):>13.4f}%"
116            f"{p2:>14.6f}{relative_error_percent(p2, theory):>13.4f}%"
117        )
118
119
120    if __name__ == "__main__":
121        main()

```

Listing A.1: Analytical verification for the hollow square waveguide.

A.2 Captured Console Output

Hollow Square Waveguide Analytical Verification

Geometry: a = 20.000 mm, b = 20.000 mm, length = 90.000 mm

Observation frequency: 10.000 GHz

Low-order analytical modes

Mode (ohm)	fc (GHz)	Status at 10 GHz	lambda_g (mm)	Z_TE
TE10 569.057	7.494811	propagating	45.284	
TE01 569.057	7.494811	propagating	45.284	
(1,1) family N/A	10.599264	evanescent	N/A	
TE20 N/A	14.989623	evanescent	N/A	
TE02 N/A	14.989623	evanescent	N/A	

Lowest-order detailed quantities

Free-space wavelength: 29.979246 mm

Guide wavelength: 45.284113 mm

TE wave impedance: 569.056941 ohm

Electrical length: 1.987452 guide wavelengths

CST cutoff comparison

Mode	Theory	Port 1	Error P1	Port 2	Error P2
1	7.494811	7.490707	-0.0548%	7.490310	-0.0601%
2	7.494811	7.491242	-0.0476%	7.490640	-0.0557%
3	10.599264	10.593500	-0.0544%	10.593100	-0.0582%

Listing A.2: Console output generated by Listing [A.1](#).

References

- [1] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: John Wiley & Sons, 2011.
- [2] W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2013.
- [3] Dassault Systèmes SIMULIA. “Cst studio suite learning edition tutorials,” Dassault Systèmes, Accessed: Jul. 25, 2026. [Online]. Available: https://3dswym.3dexperience.3ds.com/wiki/simulia-community/cst-studio-suite-learning-edition-tutorials_13_mFrIJTnmMB6fVWqzDkw.
- [4] Dassault Systèmes SIMULIA. “Microwave and rf component design,” Dassault Systèmes, Accessed: Jul. 25, 2026. [Online]. Available: <https://www.3ds.com/products/simulia/electromagnetic-simulation/rf-filters-components>.